No.	Status	Date
LNDE-RPT-20160118-004-DS-3-RF-101	Issued	2016-02-12

UT Thickness Data Sheet - 3-RF-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurements for line 3-RF-101 located within the ATU-100. The scope of work is defined by the requirements of the full-unit survey detailed in LNDE-RPT-20160118-004-R0.

Marcus Chen, acting as the lead UT technician, is responsible for the execution of the thickness measurements and the recording of data. All activities were performed in accordance with the site requirements for the Amine Treating Unit.

Technical Basis

Thickness measurements were performed under the guidelines of QSCP-NDT-001 Rev 2, incorporating the requirements for insulation removal and inspection-port access as mandated by the CUI program.

The inspection methodology conforms to the requirements of ASTM E797. For the A312 TP316 material of 3-RF-101, the instrument was configured using the stainless-steel velocity setting.

Component Design Data

The subject line, 3-RF-101, is a reflux line constructed from A312 TP316 with an outer diameter of 88.9 mm and a nominal wall thickness of 5.49 mm. The design pressure is 517 kPa(g) and the design temperature is 104 °C.

The minimum allowable thickness (t-min) for this component is 2.41 mm. Any measured value below this threshold is subject to the site NCR process.

Inspection Interval and Classification

Line 3-RF-101 is categorized as Class 2, which carries a maximum thickness measurement interval of 10 years and an external visual interval of 5 years.

The current survey, as part of LNDE-RPT-20160118-004-R0, fulfills the Class 2 10-year requirement since 2006.

Thickness readings, measured 2016-02-12

Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-101-01	7.11	3.56	2011-02-18	6.39	6.17	0.048	0.044	>50	2021-02-12	
6-SG-101-02	7.11	3.56	2011-02-18	6.14	5.76	0.065	0.076	28.8	2021-02-12	
6-SG-101-03	7.11	3.56	2011-02-18	6.17	5.87	0.061	0.060	38.1	2021-02-12	
6-SG-101-04	7.11	3.56	2011-02-18	6.05	5.58	0.076	0.094	21.4	2021-02-12	
6-SG-101-05	7.11	3.56	2011-02-18	5.87	5.38	0.088	0.098	18.5	2021-02-12	
6-SG-101-06	7.11	3.56	2011-02-18	6.27	5.88	0.060	0.078	29.6	2021-02-12	
6-SG-101-08	7.11	3.56	2011-02-18	6.33	6.00	0.066	0.066	36.8	2021-02-12	

Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-SG-102-01	7.11	3.56	2011-02-18	5.75	5.42	0.086	0.066	21.6	2021-02-12	
6-SG-102-02	7.11	3.56	2011-02-18	5.80	5.43	0.084	0.074	22.2	2021-02-12	
6-SG-102-03	7.11	3.56	2011-02-18	5.80	5.22	0.091	0.116	14.3	2021-02-12	
6-SG-102-04	7.11	3.56	2011-02-18	5.73	5.18	0.091	0.110	14.7	2021-02-12	
6-SG-102-05	7.11	3.56	2011-02-18	6.46	6.19	0.047	0.054	48.5	2021-02-12	
6-SG-102-06	7.11	3.56	2011-02-18	6.41	6.06	0.070	0.070	35.6	2021-02-12	

Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
8-TG-101-01	8.18	4.06	2006-02-24	7.89	7.76	0.022	0.013	>50	2026-02-12	
8-TG-101-02	8.18	4.06	2006-02-24	7.75	7.46	0.037	0.029	>50	2026-02-12	
8-TG-101-03	8.18	4.06	2006-02-24	7.79	7.54	0.035	0.025	>50	2026-02-12	
8-TG-101-04	8.18	4.06	2006-02-24	7.76	7.51	0.030	0.025	>50	2026-02-12	
8-TG-101-05	8.18	4.06	2006-02-24	7.95	7.73	0.025	0.022	>50	2026-02-12	
8-TG-101-07	8.18	4.06	—	—	7.39	—	—	—	2026-02-12	

Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-101-01	6.02	3.18	2006-02-24	5.51	5.02	0.049	0.049	37.4	2026-02-12	
4-LA-101-02	6.02	3.18	2006-02-24	5.76	5.45	0.028	0.031	>50	2026-02-12	
4-LA-101-03	6.02	3.18	2006-02-24	5.76	5.45	0.026	0.031	>50	2026-02-12	
4-LA-101-04	6.02	3.18	2006-02-24	5.78	5.48	0.030	0.030	>50	2026-02-12	
4-LA-101-05	6.02	3.18	2006-02-24	5.81	5.54	0.024	0.027	>50	2026-02-12	
4-LA-101-06	6.02	3.18	2006-02-24	5.64	5.30	0.038	0.034	>50	2026-02-12	
4-LA-101-07	6.02	3.18	2006-02-24	5.57	4.87	0.056	0.070	24.1	2026-02-12	
4-LA-101-09	6.02	3.18	—	—	5.00	—	—	—	2026-02-12	

Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-102-01	6.02	3.18	2006-02-24	5.63	5.25	0.039	0.038	>50	2026-02-12	
4-LA-102-02	6.02	3.18	2006-02-24	5.67	5.36	0.036	0.031	>50	2026-02-12	
4-LA-102-03	6.02	3.18	2006-02-24	5.55	5.28	0.040	0.027	>50	2026-02-12	
4-LA-102-04	6.02	3.18	2006-02-24	5.34	4.90	0.055	0.044	31.5	2026-02-12	
4-LA-102-05	6.02	3.18	2006-02-24	5.62	5.34	0.033	0.028	>50	2026-02-12	
4-LA-102-06	6.02	3.18	2006-02-24	5.83	5.60	0.020	0.023	>50	2026-02-12	
4-LA-102-07	6.02	3.18	2006-02-24	5.63	5.40	0.030	0.023	>50	2026-02-12	
4-LA-102-08	6.02	3.18	—	—	5.70	—	—	—	2026-02-12	

Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-LA-103-01	7.11	3.56	2006-02-24	6.75	6.44	0.029	0.031	>50	2026-02-12	
6-LA-103-02	7.11	3.56	2006-02-24	6.87	6.53	0.033	0.034	>50	2026-02-12	
6-LA-103-03	7.11	3.56	2006-02-24	6.65	6.28	0.040	0.037	>50	2026-02-12	
6-LA-103-04	7.11	3.56	2006-02-24	6.52	6.15	0.051	0.037	>50	2026-02-12	
6-LA-103-05	7.11	3.56	2006-02-24	6.49	5.98	0.061	0.051	39.9	2026-02-12	
6-LA-103-06	7.11	3.56	2006-02-24	6.42	5.82	0.067	0.060	33.6	2026-02-12	
6-LA-103-07	7.11	3.56	2006-02-24	6.76	6.12	0.053	0.064	39.9	2026-02-12	
6-LA-103-08	7.11	3.56	2006-02-24	6.79	6.49	0.028	0.030	>50	2026-02-12	
6-LA-103-10	7.11	3.56	—	—	6.30	—	—	—	2026-02-12	

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-01	8.56	3.81	2014-09-19	8.07	7.83	0.035	0.098	40.9	2019-02-12	
4-RA-101-02	8.56	3.81	2014-09-19	8.02	7.94	0.030	0.052	>50	2019-02-12	
4-RA-101-03	8.56	3.81	2014-09-19	7.88	7.65	0.048	0.066	>50	2019-02-12	
4-RA-101-04	8.56	3.81	2014-09-19	8.18	8.00	0.032	0.042	>50	2019-02-12	
4-RA-101-05	8.56	3.81	2014-09-19	7.98	7.82	0.039	0.068	>50	2019-02-12	
4-RA-101-06	8.56	3.81	2014-09-19	7.91	7.87	0.029	0.029	>50	2019-02-12	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-07	8.56	3.81	2014-09-19	8.22	8.14	0.023	0.032	>50	2019-02-12	
4-RA-101-08	8.56	3.81	2014-09-19	7.71	7.31	0.069	0.142	24.6	2019-02-12	
4-RA-101-09	8.56	3.81	2014-09-19	8.27	8.13	0.022	0.038	>50	2019-02-12	
4-RA-101-10	8.56	3.81	2014-09-19	8.21	8.02	0.033	0.052	>50	2019-02-12	
4-RA-101-11	8.56	3.81	2014-09-19	7.88	7.60	0.054	0.088	42.9	2019-02-12	

Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-102-01	8.56	3.81	2014-09-19	8.25	8.02	0.026	0.062	>50	2019-02-12	
4-RA-102-02	8.56	3.81	2014-09-19	7.89	7.64	0.047	0.070	>50	2019-02-12	
4-RA-102-03	8.56	3.81	2014-09-19	7.95	7.55	0.051	0.098	38.0	2019-02-12	
4-RA-102-04	8.56	3.81	2014-09-19	7.93	7.72	0.041	0.048	>50	2019-02-12	
4-RA-102-05	8.56	3.81	2014-09-19	7.84	7.67	0.044	0.080	48.1	2019-02-12	
4-RA-102-06	8.56	3.81	2014-09-19	7.95	7.73	0.039	0.056	>50	2019-02-12	
4-RA-102-07	8.56	3.81	2014-09-19	7.98	7.78	0.040	0.064	>50	2019-02-12	
4-RA-102-08	8.56	3.81	2014-09-19	7.58	7.35	0.057	0.106	33.3	2019-02-12	
4-RA-102-09	8.56	3.81	2014-09-19	7.84	7.54	0.051	0.088	42.2	2019-02-12	
4-RA-102-11	8.56	3.81	2014-09-19	8.12	7.96	0.046	0.046	>50	2019-02-12	

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-01	10.97	4.57	2014-09-19	10.62	10.29	0.036	0.036	>50	2019-02-12	
6-RA-103-02	10.97	4.57	2014-09-19	10.06	9.69	0.070	0.108	47.2	2019-02-12	
6-RA-103-03	10.97	4.57	2014-09-19	10.39	10.06	0.047	0.072	>50	2019-02-12	
6-RA-103-04	10.97	4.57	2014-09-19	9.79	9.31	0.084	0.132	35.8	2019-02-12	
6-RA-103-05	10.97	4.57	2014-09-19	9.81	9.48	0.074	0.124	39.5	2019-02-12	
6-RA-103-06	10.97	4.57	2014-09-19	10.43	10.32	0.035	0.048	>50	2019-02-12	
6-RA-103-07	10.97	4.57	2014-09-19	10.37	10.20	0.040	0.070	>50	2019-02-12	
6-RA-103-08	10.97	4.57	2014-09-19	10.49	10.33	0.032	0.022	>50	2019-02-12	
6-RA-103-09	10.97	4.57	2014-09-19	10.39	10.33	0.034	0.046	>50	2019-02-12	
6-RA-103-10	10.97	4.57	2014-09-19	9.93	9.49	0.077	0.112	43.8	2019-02-12	
6-RA-103-11	10.97	4.57	2014-09-19	10.10	9.77	0.236	0.236	22.0	2019-02-12	
6-RA-103-13	10.97	4.57	2014-09-19	10.38	10.15	0.060	0.060	>50	2019-02-12	

Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RV-101-01	7.62	3.56	2013-09-09	6.49	6.34	0.063	0.081	34.2	2019-02-12	
3-RV-101-02	7.62	3.56	2013-09-09	6.44	6.31	0.063	0.052	43.5	2019-02-12	
3-RV-101-03	7.62	3.56	2013-09-09	5.80	5.63	0.095	0.098	21.2	2019-02-12	
3-RV-101-04	7.62	3.56	2013-09-09	6.13	5.88	0.086	0.102	22.8	2019-02-12	
3-RV-101-05	7.62	3.56	2013-09-09	6.48	6.36	0.066	0.050	42.6	2019-02-12	
3-RV-101-06	7.62	3.56	2013-09-09	6.11	5.96	0.082	0.087	27.7	2019-02-12	
3-RV-101-07	7.62	3.56	2013-09-09	5.73	5.54	0.122	0.122	16.2	2019-02-12	
3-RV-101-08	7.62	3.56	2013-09-09	6.36	6.31	0.074	0.074	37.2	2019-02-12	

Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-01	5.54	2.79	2013-09-09	4.49	4.35	0.059	0.037	26.4	2019-02-12	
2-RV-102-02	5.54	2.79	2013-09-09	4.29	4.01	0.079	0.109	11.2	2019-02-12	
2-RV-102-03	5.54	2.79	2013-09-09	4.68	4.64	0.042	0.037	43.9	2019-02-12	

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RV-102-04	5.54	2.79	2013-09-09	4.81	4.73	0.043	0.020	45.5	2019-02-12	
2-RV-102-05	5.54	2.79	2013-09-09	4.96	4.97	0.013	0.013	>50	2019-02-12	
2-RV-102-06	5.54	2.79	2013-09-09	4.55	4.27	0.102	0.102	14.6	2019-02-12	

Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-RF-101-01	5.49	2.41	2013-12-09	5.37	5.31	0.009	0.017	>50	2026-02-12	
3-RF-101-02	5.49	2.41	2013-12-09	5.40	5.34	0.006	0.003	>50	2026-02-12	
3-RF-101-03	5.49	2.41	2013-12-09	5.37	5.33	0.006	0.012	>50	2026-02-12	
3-RF-101-04	5.49	2.41	2013-12-09	5.41	5.35	0.006	-0.002	>50	2026-02-12	
3-RF-101-05	5.49	2.41	2013-12-09	5.32	5.33	0.006	0.007	>50	2026-02-12	
3-RF-101-06	5.49	2.41	2013-12-09	5.17	5.23	0.015	0.017	>50	2026-02-12	
3-RF-101-07	5.49	2.41	2013-12-09	5.48	5.37	0.003	-0.002	>50	2026-02-12	
3-RF-101-09	5.49	2.41	2013-12-09	5.26	5.19	-0.029	-0.029	>50	2026-02-12	

Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-RF-102-01	3.91	2.16	2013-12-09	3.60	3.76	0.008	0.009	>50	2026-02-12	
2-RF-102-02	3.91	2.16	2013-12-09	3.70	3.64	0.012	0.024	>50	2026-02-12	
2-RF-102-03	3.91	2.16	2013-12-09	3.79	3.75	0.009	0.005	>50	2026-02-12	
2-RF-102-04	3.91	2.16	2013-12-09	3.82	3.38	0.025	0.043	28.3	2026-02-12	
2-RF-102-05	3.91	2.16	2013-12-09	3.67	3.49	0.082	0.082	16.1	2024-03-06	
2-RF-102-06	3.91	2.16	2013-12-09	3.67	3.47	0.099	0.099	13.2	2022-09-26	
2-RF-102-07	3.91	2.16	2013-12-09	3.72	3.57	0.020	0.028	>50	2026-02-12	

Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-LA-104-01	6.02	3.18	2006-02-24	5.75	5.31	0.038	0.044	48.2	2026-02-12	
4-LA-104-02	6.02	3.18	2006-02-24	5.56	5.00	0.052	0.056	32.4	2026-02-12	
4-LA-104-03	6.02	3.18	2006-02-24	5.74	5.42	0.027	0.032	>50	2026-02-12	
4-LA-104-04	6.02	3.18	2006-02-24	5.86	5.63	0.022	0.023	>50	2026-02-12	
4-LA-104-05	6.02	3.18	2006-02-24	5.51	4.88	0.058	0.063	26.9	2026-02-12	
4-LA-104-06	6.02	3.18	2006-02-24	5.83	5.55	0.028	0.028	>50	2026-02-12	
4-LA-104-07	6.02	3.18	2006-02-24	5.62	5.28	0.037	0.034	>50	2026-02-12	
4-LA-104-08	6.02	3.18	2006-02-24	5.50	5.11	0.042	0.039	46.3	2026-02-12	
4-LA-104-10	6.02	3.18	—	—	4.84	—	—	—	2026-02-12	

Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-104-01	8.56	3.81	2014-09-19	7.84	7.71	0.040	0.048	>50	2019-02-12	
4-RA-104-02	8.56	3.81	2014-09-19	8.09	7.85	0.035	0.058	>50	2019-02-12	
4-RA-104-03	8.56	3.81	2014-09-19	8.46	8.28	0.057	0.056	>50	2019-02-12	
4-RA-104-04	8.56	3.81	2014-09-19	8.49	8.18	0.063	0.084	>50	2019-02-12	
4-RA-104-05	8.56	3.81	2014-09-19	8.37	8.16	0.050	0.082	>50	2019-02-12	
4-RA-104-06	8.56	3.81	2014-09-19	7.91	7.61	0.049	0.074	>50	2019-02-12	
4-RA-104-07	8.56	3.81	2014-09-19	7.61	7.32	0.064	0.092	38.0	2019-02-12	
4-RA-104-08	8.56	3.81	2014-09-19	8.00	7.81	0.074	0.074	>50	2019-02-12	

Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
2-ST-101-01	3.91	2.29	2006-02-24	3.64	3.61	0.016	0.003	>50	2026-02-12	
2-ST-101-02	3.91	2.29	2006-02-24	3.67	3.55	0.018	0.012	>50	2026-02-12	
2-ST-101-03	3.91	2.29	2006-02-24	3.82	3.55	0.020	0.027	46.5	2026-02-12	
2-ST-101-04	3.91	2.29	2006-02-24	3.77	3.68	0.007	0.009	>50	2026-02-12	
2-ST-101-05	3.91	2.29	2006-02-24	3.72	3.50	0.021	0.022	>50	2026-02-12	
2-ST-101-06	3.91	2.29	—	—	3.63	—	—	—	2026-02-12	

Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-101-01	5.49	2.79	2006-02-24	5.13	5.03	0.020	0.010	>50	2026-02-12	
3-CW-101-02	5.49	2.79	2006-02-24	5.20	5.01	0.020	0.019	>50	2026-02-12	
3-CW-101-03	5.49	2.79	2006-02-24	5.08	4.93	0.030	0.015	>50	2026-02-12	
3-CW-101-04	5.49	2.79	2006-02-24	5.38	5.26	0.010	0.012	>50	2026-02-12	
3-CW-101-05	5.49	2.79	2006-02-24	5.33	5.35	0.007	-0.002	>50	2026-02-12	
3-CW-101-06	5.49	2.79	—	—	5.08	—	—	—	2026-02-12	

Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
3-CW-102-01	5.49	2.79	2006-02-24	5.20	5.06	0.022	0.014	>50	2026-02-12	
3-CW-102-02	5.49	2.79	2006-02-24	5.02	4.37	0.057	0.065	24.2	2026-02-12	
3-CW-102-03	5.49	2.79	2006-02-24	5.07	4.54	0.050	0.053	32.9	2026-02-12	
3-CW-102-04	5.49	2.79	2006-02-24	5.03	4.69	0.041	0.034	46.2	2026-02-12	
3-CW-102-05	5.49	2.79	—	—	4.66	—	—	—	2026-02-12	

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT Level II; CAN/CGSB-48.9712 UT Level II	2016-02-12	_____